

ECON 712: Applied Econometrics

Lecture 5: Linear Regression using Vector Spaces

Vector Spaces

- An interesting way to understand regression analysis is to study it using vector spaces
- There are three concepts in vector geometry that are used to study OLS:
 1. Scalar product
 2. Vector Addition
 3. Subspaces of Euclidean Spaces

Scalar Product

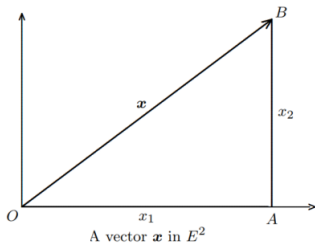
- Scalar product is a single number that determines the relationship between two vectors of the same dimension:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$$

where both vectors are of size $n \times 1$

- An important property application of the scalar product is the **length** or **norm** of a vector defined as:

$$\|\mathbf{x}\| = \left[\sum_{i=1}^n x_i^2 \right]^{\frac{1}{2}} = \langle \mathbf{x}, \mathbf{x} \rangle^{\frac{1}{2}}$$



Scalar Product

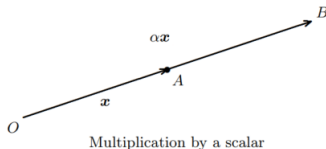
- Scalar product is a single number that determines the relationship between two vectors of the same dimension:

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where both vectors are of size $n \times 1$

- Another important property is that if the angle between two vectors, θ , is zero, then:
 - ▶ This property is based on multiplying a vector by a scalar:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \|\mathbf{x}\| \|\mathbf{y}\|$$



Scalar Product

- Scalar product is a single number that determines the relationship between two vectors of the same dimension:

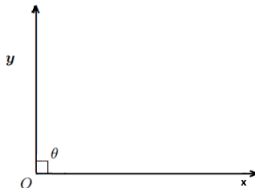
$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T \mathbf{y} = \sum_{i=1}^n x_i y_i$$

where both vectors are of size $n \times 1$

- Lastly if the angle between two vectors, $\theta = 90^\circ$ (i.e. they are **orthogonal**), then:

► E.g. $\mathbf{x}^T = [1, 0]$, $\mathbf{y}^T = [0, 1]$

$$\langle \mathbf{x}, \mathbf{y} \rangle = 0$$

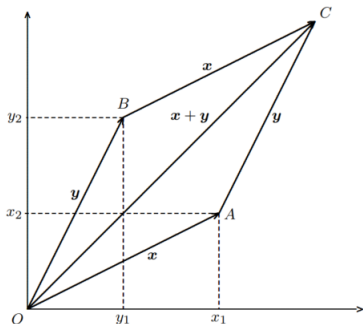


Vector Addition

- Adding two vectors of size n results in:

$$\mathbf{z} = \mathbf{x} + \mathbf{y} = \begin{bmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{bmatrix}$$

Vector addition in 2 dimensions



- Vector addition can be combined with multiplication by a scalar

Euclidean Spaces and Subspaces

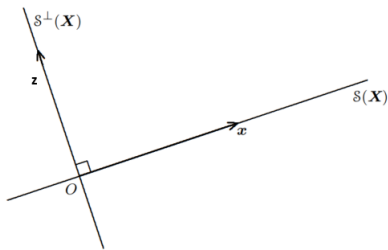
- Vectors of size n can be thought as coordinates of a point in E^n
- The minimum set of vectors that spans a particular space are called the **basis** vectors
 - ▶ We need k linearly independent vectors of size $n \geq k$ to span the k -dimensional space
 - ▶ If $k < n$, we call k a subspace of E^n

Orthogonal Complement

- We call an orthogonal complement, the set of vectors in E^n that are orthogonal to the basis vectors of a k -dimensional subspace.

Example with $n = 2$, $k = 1$

$$\langle \mathbf{x}, \mathbf{z} \rangle = 0$$



The spaces $S(\mathbf{X})$ and $S^\perp(\mathbf{X})$

- These vectors live in the $n - k$ space

Vector Space Definitions

- We use set of vectors to formally define the following subspaces:

1. k -dimensional subspace:

$$\mathcal{S}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k) \equiv \left\{ \mathbf{z} \in E^n \mid \mathbf{z} = \sum_{i=1}^k b_i \mathbf{x}_i, b_i \in \mathbb{R} \right\}$$

2. $(n - k)$ -dimensional orthogonal complement:

$$\mathcal{S}^\perp(\mathbf{w}) \equiv \{ \mathbf{w} \in E^n \mid \mathbf{w}^T \mathbf{z} = 0 \text{ for all } \mathbf{z} \in \mathcal{S}(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k) \}$$

- We will also make use of the following inequality, called Cauchy-Schwartz:

$$\langle \mathbf{x}, \mathbf{y} \rangle \leq \|\mathbf{x}\| \|\mathbf{y}\|$$

Linear Regression

- Linear Regression considers each linearly-independent k column vector of size n from $\mathbf{X}$ as the basis vector that spans the k -dimensional space

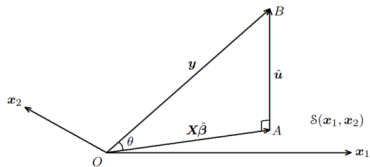
- ▶ We also assume the following two conditions:

$$\mathbf{X}^T \hat{\mathbf{u}} = \mathbf{0} \quad , \quad \mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}$$

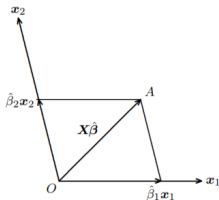
- ▶ This results in linear regression being defined as selecting $\hat{\boldsymbol{\beta}}$ that reaches the perpendicular dropped from $\mathbf{y}$ vector into the plane spanned by $\mathbf{X}$

Linear Regression

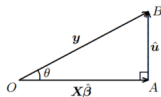
Example with $n = 3$, $k = 2$



(a) y projected on two regressors



(b) The span $S(x_1, x_2)$ of the regressors



(c) The vertical plane through y

Linear regression in three dimensions

Linear Regression

- The concept of dropping a perpendicular is formalized by an orthogonal projection:

$$P_x = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$$

- ▶ An orthogonal projection is a matrix that projects any vector to a point in the subspace $\mathcal{S}(\mathbf{X})$ that it is closest to it
- ▶ $\mathcal{S}(\mathbf{X})$ is called the invariant subspace of P_x because the projection of any vector in $\mathcal{S}(\mathbf{X})$ using P_x leaves that vector unchanged
- We are also going to use a projection onto the orthogonal subspace of $\mathcal{S}(\mathbf{X})$, $\mathcal{S}^\perp$

$$M_x = I - P_x$$